

Reflection Frank van Harmelen: There are 2 main lines of history/development in AI

- Symbolical representations (knowledge, logic, reasoning) Autonomous planning, reasoning, diagnosis, decision support, language generation, search engines
- Statistical representations (connection, data, learning, statistics) Pattern recognition, motor skills, speech generation, search engines

Future of AI has to come from a combination of these 2.

He talks about strength & weaknesses of human experts versus machine learning. Machines need a lot of data that is not always available and its expensive. Machines can't really explain themselves.

Our brain is able to do both slow thinking and fast thinking. The question is how we are going to combine the strength of the 2 (human brain and machine)

He also explains what knowledge representation is. Reasoning about the world, making a model of the world. A representation of a thing is not the thing itself.

Some questions we don't have an answer to yet:

- -Why are some networks connected and some are not (why do many nodes connect to wikipedia.. Because that has the best knowledge, or the most content?)
- -Why are some nodes close to each other and some far away (does the path length say anything about the semantic distance?)
- -Do networks with more nodes have more influence than those with low-level structures.
- -How explosive is inference (small facts bases vs large facts (small is 10m facts:)), how fast does a KB grow, and why are their differences between knowledge bases
- -Why can you answer 95% of the queries with 5% of the knowledge base (hotspot)?

Thoughts..

Very inspiring person, enthusiastic, expert and pioneer in the field. I liked how he raised the philosophical issues we should all think about. I would really like to be a part of this research someday:)

My personal take outs:

-Find out more about hybrid intelligence and stay updated. I didn't really understand what it is exactly, so I'll find out. What will VU's research will be about, what it's goal is and how this fund is going to be allocated. Maybe I could be a part of it somehow..

- -Get familiar with semantic web, basic principles and what it could lead to, how it works
- -Always look for the combination of both symbolical and statistical representations.